

Probability with Combinatorics

Find the probability of each event.

- 1) Beth and Shayna each purchase one raffle ticket. If a total of eleven raffle tickets are sold and two winners will be selected, what is the probability that both Beth and Shayna win?
- 2) A meeting takes place between a diplomat and fourteen government officials. However, four of the officials are actually spies. If the diplomat gives secret information to ten of the attendees at random, what is the probability that no secret information was given to the spies?
- 3) A fair coin is flipped ten times. What is the probability of the coin landing heads up exactly six times?
- 4) A six-sided die is rolled six times. What is the probability that the die will show an even number exactly two times?
- 5) A test consists of nine true/false questions. A student who forgot to study guesses randomly on every question. What is the probability that the student answers at least two questions correctly?
- 6) A basketball player has a 50% chance of making each free throw. What is the probability that the player makes at least eleven out of twelve free throws?